

SafeDig AI Map QA & Validation Platform

Comprehensive Technology Stack & AI Architecture Specification

1. Architecture Overview

SafeDig employs a dual-engine architecture designed specifically for safety-critical utility validation:

1. Deterministic Core: PyMuPDF vector math, Shapely 2D spatial geometry, OpenCV computer vision, and a 17-Gate Policy Engine. This layer delivers 100% auditable, zero-hallucination verification.
2. AI / LLM Advisory Layer: LangGraph pipeline orchestration paired with local on-premise LLMs (Qwen 2.5 / Llama 3.2) providing natural language explanations and discrepancy summaries in an advisory capacity only.

2. Backend & Core Server Framework

Component	Technology / Library	Version	Role & Purpose in Platform
Core Language	Python	3.11+	High-speed typing, pattern matching, async core
REST API	FastAPI	>=0.100.0	Asynchronous endpoints, OpenAPI generation
ASGI Web Server	Uvicorn (Standard)	>=0.22.0	High-throughput non-blocking production server
Type Validation	Pydantic & Settings	>=2.0.0	Strict domain type enforcement and env parsing
Async Networking	HTTPX & AnyIO	>=0.25.0	Non-blocking client calls and concurrent I/O

3. Computer Vision, Spatial Math & Vector Extraction

Component	Technology / Library	Version	Role & Purpose in Platform
PDF Engine	PyMuPDF (fitz)	>=1.24.0	Vector path extraction, dash analysis, 300 DPI rasterizer
Spatial Geometry	Shapely	>=2.0.0	Exact 2D polygon intersection, AOI bounding & buffer math
Computer Vision	OpenCV (headless)	>=4.8.0	Raster filtering, symbol contour analysis, crop extraction
Data Parsing	Pandas	>=2.0.0	Upstream enquiry table indexing and record normalization
Spreadsheets	OpenPyXL	>=3.1.0	High-fidelity .xlsx parsing for utility disclosure sheets
Persistence	SQLAlchemy +aiosqlite	>=2.0.0	Async SQLite (safedig.db) & PostgreSQL production support

4. AI Orchestration, Local LLMs & Advisory Models

AI Module	Framework / Model	Execution Host	Role & Safety Authority
Workflow Graph	LangGraph (>=0.2.0)	In-Process DAG	Stateful deterministic pipeline orchestration
Agent Abstraction	LangChain-Core (>=0.3.0)	In-Process Core	Standardized messages, prompts & agent states
Primary LLM	Qwen 2.5 (7B / 14B / Latest)	Ollama (On-Premise)	Technical reasoning, discrepancy explanation
Alternative LLM	Llama 3.2 (3B) / 3.1 (8B)	Ollama (On-Premise)	Low-resource edge / fast advisory generation
Cloud Models	Claude 3.5 Sonnet / GPT-4o	Cloud API (Optional)	Configurable cloud copilot via settings API
Rule Fallback	Deterministic Explainer	Internal Python	0ms offline fallback if LLM is unavailable

MANDATORY SAFETY PRINCIPLE: ZERO VALIDATION AUTHORITY FOR LLM

Under SafeDig enterprise policy, the LLM has ADVISORY AUTHORITY ONLY. No safety gate, auto-clear decision, or hazard finding can be overridden by model inferences. If the LLM produces a contradiction or becomes unresponsive, the system automatically defaults to deterministic safety rules with zero latency.

5. Frontend Web Console & Inspection Engine

Design Framework	Tailwind CSS	Utility-first responsive framework, custom dark theme with high-contrast accents
Typography	Google Font Inter	400 to 800 weights with fluid CSS clamp() scaling for all viewports
Vector Icons	Lucide Icons	Lightweight SVG icon library for status badges, controls, and navigation
Inspection Engine	DOM / Canvas Zoom	Hardware-accelerated pan (drag) & zoom (50% to 600%), AOI centering & fullscreen

6. Automated Quality Assurance & Verification Suite

Test Runner	PyTest (>=8.0.0)	75 automated unit and integration tests covering all pipeline modules
Async Tests	pytest-asyncio	Full async testing of FastAPI endpoints, async SQLite queries & state machine
Coverage	pytest-cov	Regression tracking across AOI, Vector Extraction, CV Engine, and Policy Gates